Infrastructure Gap

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Preliminary: Read in the data

Here I load the dataset

```
mukoki_gap_data <- readxl::read_xlsx("dataset_40_countries.xlsx") %>%

  janitor::clean_names() %>%

  data.frame()
```

Dependent and Independent Variables

I select the dependent and independent variables

```
modelling_data <- mukoki_gap_data %>%

  select(countries, country, state, year,

         income_group_2009, income_2009,

         overall_index, infra_index,

         transport_gap, electricity_gap,

         ict_gap, water_gap, overall_index,

         transport_invest, electricity_invest,

         ict_invest, water_invest, water_invest,

         stock_mkt_cap, human_di, exchange_rate,

         rinterestrate, percent_change_in_gfcf_public,

         fdi_gdp, govenance_index) %>%

  tibble()
```

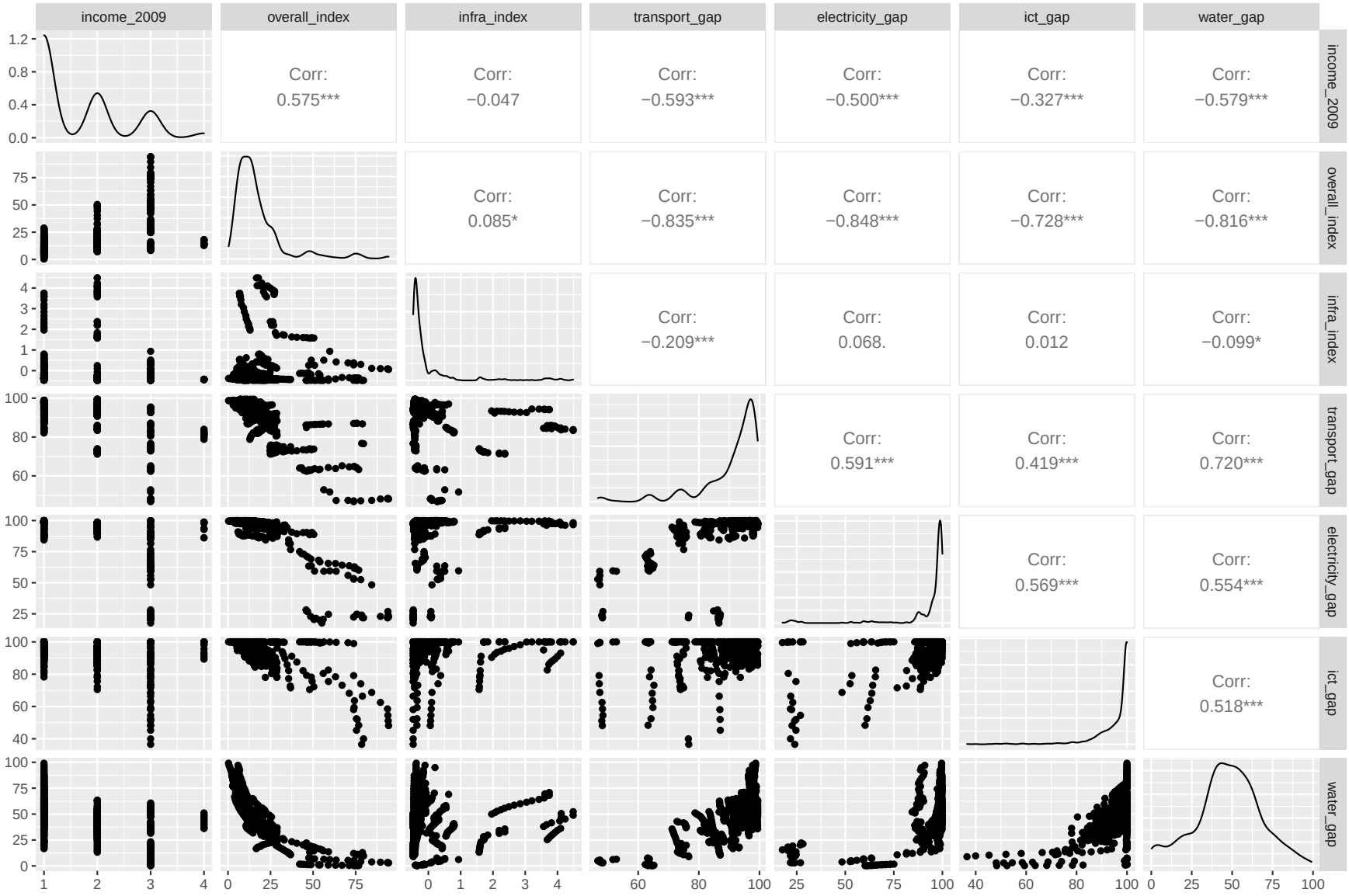
Table 1: Independent Variables

skim_variable	numeric.mean	numeric.sd	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100
income_2009	1.62500	0.8279167	1.0000000	1.0000000	1.0000000	2.0000000	4.000000
overall_index	17.86327	15.9435392	0.3687802	8.4396914	13.4978230	20.3022817	94.323663
infra_index	0.00000	0.9085647	-0.4701808	-0.4064351	-0.3251464	-0.1158849	4.485926
transport_gap	90.59428	10.2512712	46.6914350	87.9674335	94.2978910	97.3884095	99.445572
electricity_gap	92.88688	15.0433290	17.6244140	94.7152030	98.0558580	99.1708688	100.000000
ict_gap	95.26291	8.7731510	36.5554970	93.8749290	99.5557825	99.9965885	99.999990
water_gap	47.30374	20.1042116	0.2118731	36.6460240	48.1691050	60.4871940	99.109127

Dependent Variables

```
modelling_data %>%
  select(income_2009,
         overall_index, infra_index,
         transport_gap, electricity_gap,
         ict_gap, water_gap) %>%
  na.omit() %>%
  skim_without_charts() %>%
  select(-n_missing, -complete_rate, -skim_type) %>%
  kbl(., booktabs = TRUE, caption = "Independent Variables") %>%
  kable_classic(full_width = TRUE)
```

```
modelling_data %>%
  select(income_2009,
         overall_index, infra_index,
         transport_gap, electricity_gap,
         ict_gap, water_gap) %>%
  na.omit() %>%
  ggpairs()
```


income_2009
overall_index
infra_index
transport_gap
electricity_gap
ict_gap
water_gap
Corr: 0.575***
Corr: −0.047
Corr: −0.593***
Corr: −0.500***
Corr: −0.327***
Corr: −0.579***
Corr: 0.085*
Corr: −0.835***
Corr: −0.848***
Corr: −0.728***
Corr: −0.816***
Corr: −0.209***
Corr: 0.068.
Corr: 0.012
Corr: −0.099*
Corr: 0.591***
Corr: 0.419***
Corr: 0.720***
Corr: 0.569***
Corr: 0.554***
Corr: 0.518***

Independent Variables

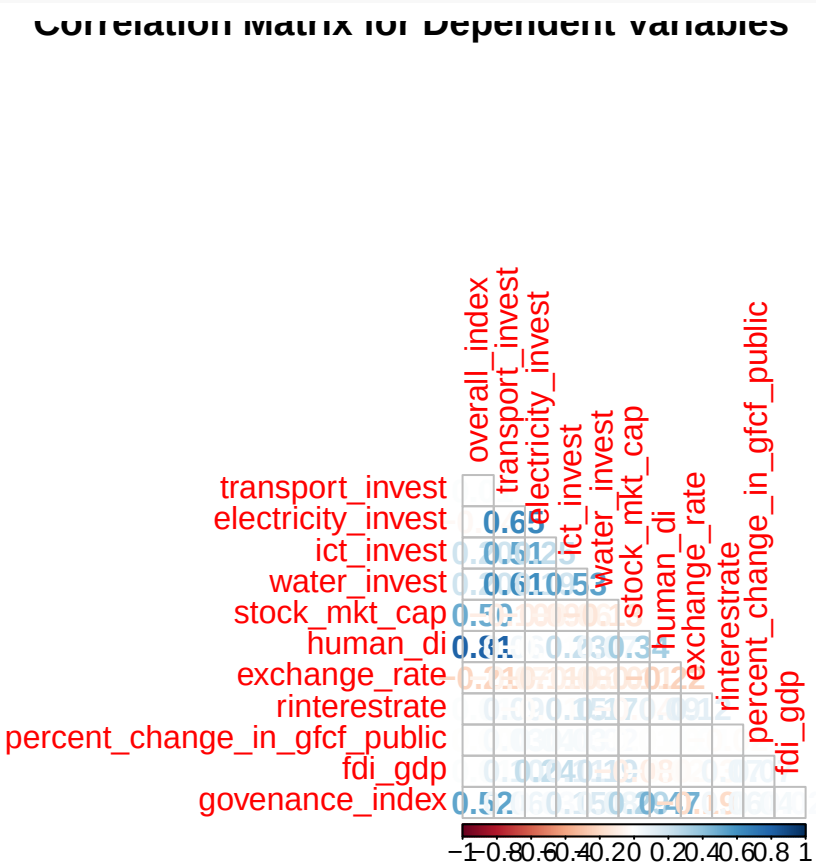
We examine the independent variables. Please confirm that these are correct before I go ahead and run the regressions. I am especially keen on overall index and its meaning. What variable in the data stands for Public-Private Partnerships (PPP) investment in Aggregate/overall infrastructure? I dont think its the overall index.

```
modelling_data %>%

  select(overall_index,

         transport_invest, electricity_invest,

         ict_invest, water_invest, water_invest,

         stock_mkt_cap, human_di, exchange_rate,

         rinterestrate, percent_change_in_gfcf_public,

         fdi_gdp, govenance_index) %>%

  skim_without_charts() %>%

  select(-n_missing, -complete_rate, -skim_type) %>%

  kbl(., booktabs = TRUE, caption = "Independent Variables") %>%

  kable_classic(full_width = TRUE)
```

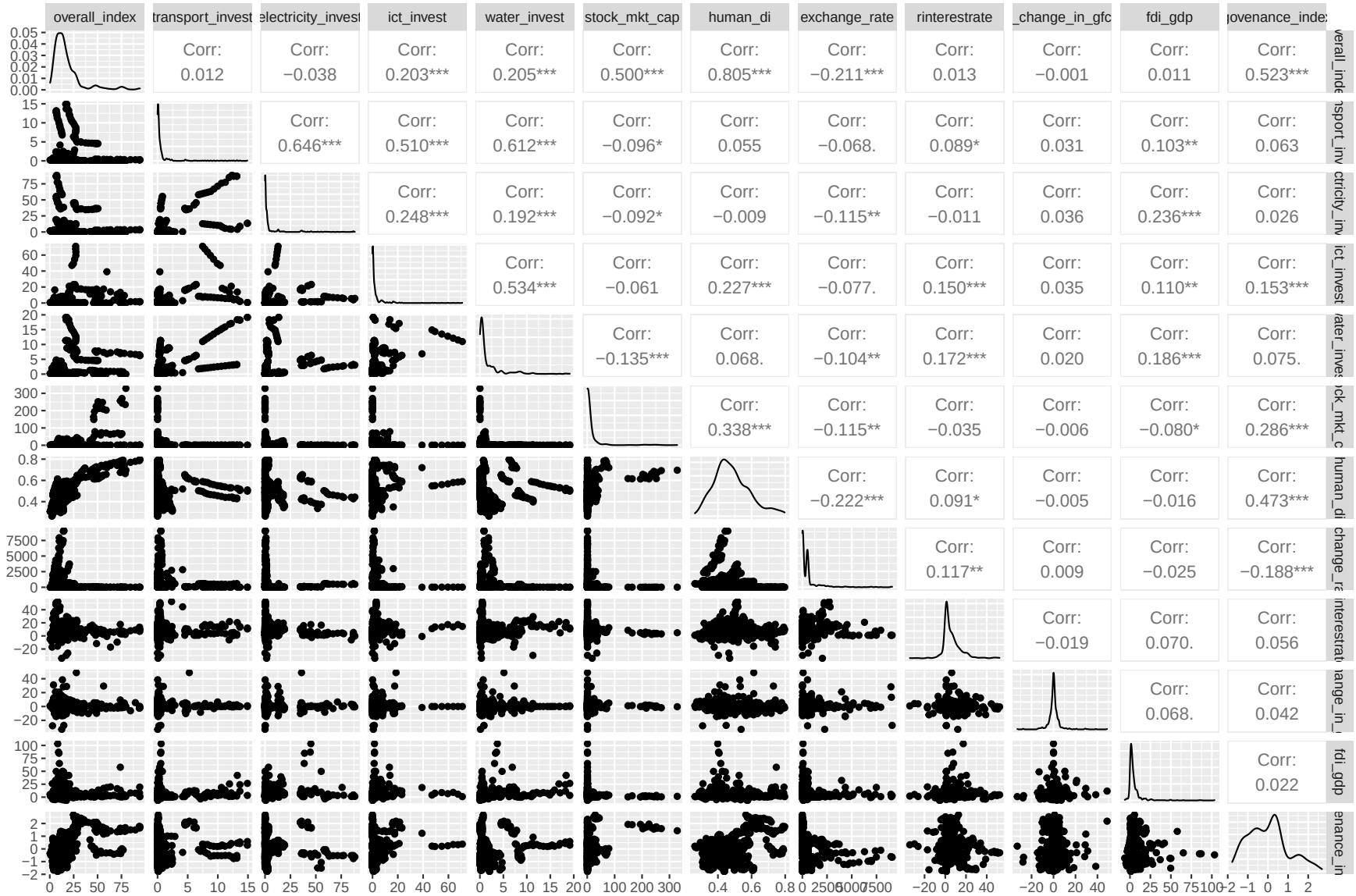
Table 2: Independent Variables

skim_variable	numeric.mean	numeric.sd	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100
overall_index	17.8632724	15.9435392	0.3687802	8.4396914	13.4978230	20.3022817	94.323663
transport_invest	0.9829726	2.4573516	0.0000908	0.0520468	0.1884312	0.5139178	14.905687
electricity_invest	5.6575235	14.3728215	0.0013178	0.1495541	0.6089139	2.3695731	88.429515
ict_invest	2.8558644	7.2741104	0.0001807	0.1146541	0.6695169	2.1571728	70.985354
water_invest	1.9211378	3.1850498	0.0146976	0.2785261	0.5765471	2.0650996	19.154820
stock_mkt_cap	10.2802022	36.0435167	0.7241040	1.0000000	1.0000000	1.0000000	328.361000
human_di	0.4838609	0.1074758	0.2630000	0.4120000	0.4680000	0.5485000	0.797000
exchange_rate	652.2826577	1260.5619533	0.8667643	17.6843045	370.4251100	555.4500000	9088.320000
rinterestrate	6.1573402	9.0830600	-34.2103090	1.0000000	3.8272198	9.2988998	52.436821
percent_change_in_gfcf	0.3170261	5.5788054	-33.0171620	-1.3669003	0.0000000	1.5717645	48.789712
fdi_gdp	4.9825751	9.3207813	-6.3698773	1.0000000	2.5635350	5.3448228	103.337390
govenance_index	0.0000000	0.9842590	-1.7782380	-0.7445558	-0.0166953	0.4886982	2.688182

```r
modelling_data %>%
  select(overall_index,
         transport_invest, electricity_invest,
         ict_invest, water_invest, water_invest,
         stock_mkt_cap, human_di, exchange_rate,
         rinterestrate, percent_change_in_gfcf_public,
         fdi_gdp, govenance_index) %>%
  drop_na(percent_change_in_gfcf_public) %>%
  cor() %>%
  corrplot(method = "number", type = "lower",
           diag = FALSE, number.digits = 2,
           title = "Correlation Matrix for Dependent Variables")
```



```
modelling_data %>%
  select(overall_index,
         transport_invest, electricity_invest,
         ict_invest, water_invest, water_invest,
         stock_mkt_cap, human_di, exchange_rate,
         rinterestrate, percent_change_in_gfcf_public,
         fdi_gdp, govenance_index) %>%
  drop_na(percent_change_in_gfcf_public) %>%
  ggpairs()
```


overall_index
transport_invest
electricity_invest
ict_invest
water_invest
stock_mkt_cap
human_di
exchange_rate
rinterestrate
_change_in_gfc
fdi_gdp
governance_index
Corr: 0.012
Corr: −0.038
Corr: 0.203***
Corr: 0.205***
Corr: 0.500***
Corr: 0.805***
Corr: −0.211***
Corr: 0.013
Corr: −0.001
Corr: 0.011
Corr: 0.523***
Corr: 0.646***
Corr: 0.510***
Corr: 0.612***
Corr: −0.096*
Corr: 0.055
Corr: −0.068.
Corr: 0.089*
Corr: 0.031
Corr: 0.103**
Corr: 0.063
Corr: 0.248***
Corr: 0.192***
Corr: −0.092*
Corr: −0.009
Corr: −0.115**
Corr: −0.011
Corr: 0.036
Corr: 0.236***
Corr: 0.026
Corr: 0.534***
Corr: −0.061
Corr: 0.227***
Corr: −0.077.
Corr: 0.150***
Corr: 0.035
Corr: 0.110**
Corr: 0.153***
Corr: −0.135***
Corr: 0.068.
Corr: −0.104**
Corr: 0.172***
Corr: 0.020
Corr: 0.186***
Corr: 0.075.
Corr: 0.338***
Corr: −0.115**
Corr: −0.035
Corr: −0.006
Corr: −0.080*
Corr: 0.286***
Corr: −0.222***
Corr: 0.091*
Corr: −0.005
Corr: −0.016
Corr: 0.473***
Corr: 0.117**
Corr: 0.009
Corr: −0.025
Corr: −0.188***
Corr: −0.019
Corr: 0.070.
Corr: 0.056
Corr: 0.068.
Corr: 0.042
Corr: 0.022

Write a regression function to run fixed and random effects models